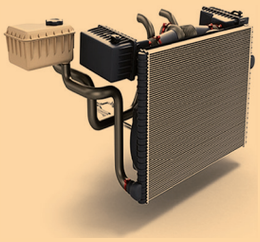


What is a Radiator?

Radiators are heat exchangers used to transfer thermal energy from one medium to another for the purpose of cooling and heating.

In automobiles it is responsible for preventing the car engine from overheating.

It uses coolant (water or oil) to keep the engine running at a healthy temperature.



7.3.1(a) Tubular Type Radiator

Small diameter tubes are used for connecting the upper and lower tank in the radiator. Water gets cooled by flowing through the number of tubes. The outside of the tube walls are attached with cooling fins. The fins and tubes are made up of pure copper. The fins are arranged horizontally with each other. In this type of radiators the cooling process is affected for the entire distance of tube if there is any blockage inside the tubes. This type of radiator is lower in weight and simple in structure and hence used in most of the vehicles.

7.3.1(b) Cellular (or) Honeycomb Type Radiator

In this type of radiator, the hot water coming from the upper tank is allowed to pass through the tubes which are in honeycombed structure and gets cooled before reaching the lower tank. Two honeycombed structure tubes are connected in between the water flowing passage. By this structure, any block occurs inside the tube will not reduce the cooling performance. This type of radiator is mostly used in racing cars. They are costlier than tube type radiators. The figure 7.3.1(b)shows the picture of the honeycomb radiator.



Figure 7.3.1(a) Tubular type radiator

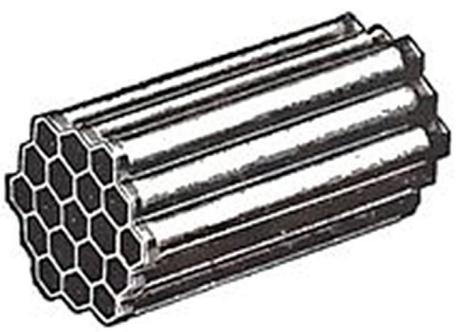


Fig 7.3.1(b) Honey comb type radiator

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